MASTER'S THESIS DEFENSE

Emotions in Perfume Reviews

Advanced Sentiment Analysis Framework for Olfactory Perception Mining: Utilizing Machine Learning and Deep Learning Techniques on Social Media Data Streams

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Why look beyond positive and negative reviews?

Perfume reviews describe sensory impressions and emotional associations.

Enjoyment

“This makes me happy.”

Anticipation

“I cannot wait to wear this.”

Two positive reviews; different emotional meanings. Illustrative examples.

Emotion classification offers a more detailed description of review language.

Nikitina & Furuoka (2023); Nandwani & Verma (2021).

One review can express several emotions

> **“This perfume makes me happy, and I cannot wait to wear it.”**

Illustrative example

Joy + **Anticipation**

The task allows multiple emotion labels for the same perfume, based on its review text.

Joy | **Trust** | **Anticipation** | **Surprise** | **Sadness** | **Fear** | **Disgust**

Target: emotion expressed in review language. Wearer responses are not directly measured.

Existing methods provide different kinds of evidence

Approach	What it contributes
Emotion dictionaries	Associations between words and emotions
Affect dimensions	Pleasantness, activation and control
Contextual language models	Patterns that depend on surrounding text

This thesis evaluates these approaches within a shared perfume-review benchmark.

Mohammad & Turney (2013); Mohammad (2018); Demszky et al. (2020); Devlin et al. (2019).

Three research questions

RQ1

Representation

How can perfume-review transcripts be represented as structured emotion features?

RQ2

Model comparison

How do machine learning and deep learning methods compare in identifying multiple emotions?

RQ3

Synthetic augmentation

How does weighted augmentation with generated reviews affect prediction on held-out YouTube perfumes?

What this thesis contributes

A perfume-review dataset

English YouTube transcript records linked to 784 perfumes.

A reproducible workflow

A 58-feature representation alongside a direct text modelling route.

A controlled comparison

Five models, feature combinations and weighted synthetic augmentation.

Shared evaluation conditions connect the dataset, representations and experiments.

From review transcripts to perfume-level predictions

784
perfumes

4,881
usable transcript records

1–10
records per perfume; median 7

Perfume-specific YouTube searches supplied available English transcripts.

Retained transcripts contain at least 20 words.

Transcript records → **Grouped by perfume** → **Emotion predictions for each perfume**

Perfumes can share source material, so the records are not necessarily independent.

Thesis §3.2.1; Table 1.

The reference labels follow a fixed annotation policy

Gemini 3.1 Pro

Assigned emotion labels from supplied review excerpts.

Fixed instructions and a consistent label set.

What the model saw

One or two excerpts per perfume.

Maximum 360 characters per excerpt.

Evaluation measures agreement with these model-generated reference labels.

Independent human validation was not conducted.

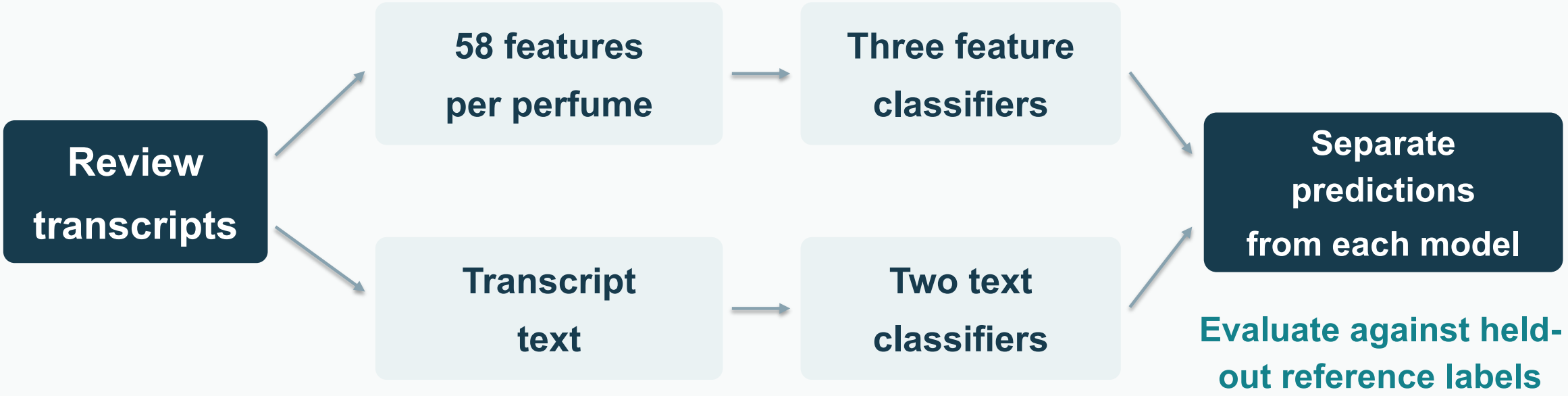
Thesis §3.4; Niu et al. (2024), methodological context.

The AI tools have four different roles

Component	Role
Pretrained text models	Produce numerical input features
Gemini 3.1 Pro	Assign reference labels
Five supervised classifiers	Learn to predict the emotion labels
GPT-5	Generate additional review text for augmentation

Feature generation, annotation, prediction and text generation are separate steps.

Two routes lead to the same prediction task



Reference labels are training targets. Held-out predictions use only model inputs.

Simplified from thesis Figure 1; §§3.5–3.7.

What do the 58 features describe?

Feature family	Count	Meaning
Emotion and sentiment word counts	10	Vocabulary associated with emotions and sentiment
Affect dimensions	3	Pleasantness, activation and control
Pretrained model scores	45	Signals from existing text classifiers
Total	**58**	**One numerical representation per perfume**

The representation combines dictionary evidence with contextual model outputs.

Thesis §3.5; Table 5.

Five models, grouped by their inputs

58 structured features

Logistic Regression

XGBoost

Random Forest

Learn from the 58-value representation.

Review text

BERT

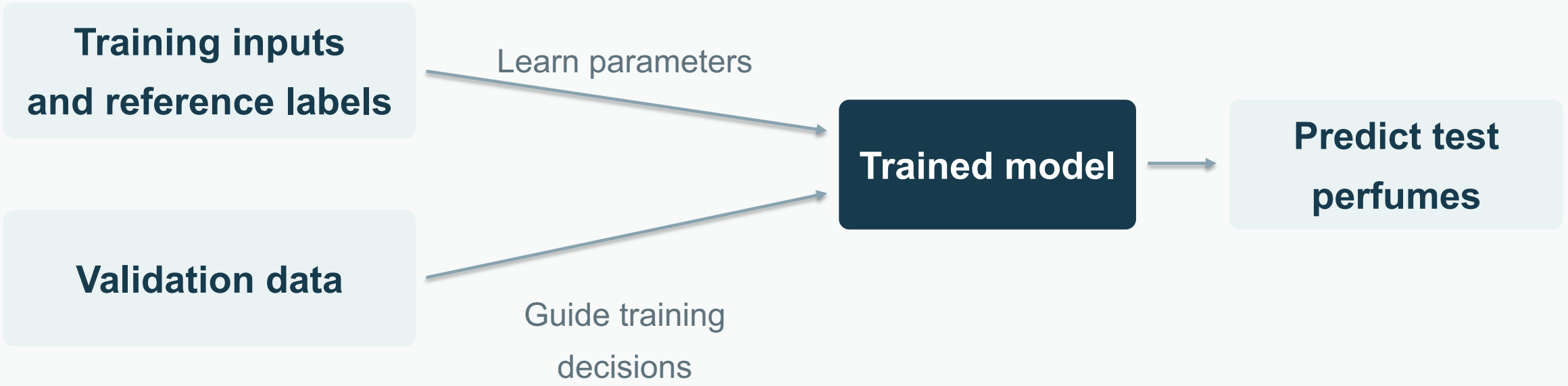
BiLSTM with Attention

Learn from transcripts; combine predictions at perfume level.

The comparison covers complete approaches, including their different inputs.

Thesis §3.6; Table 6.

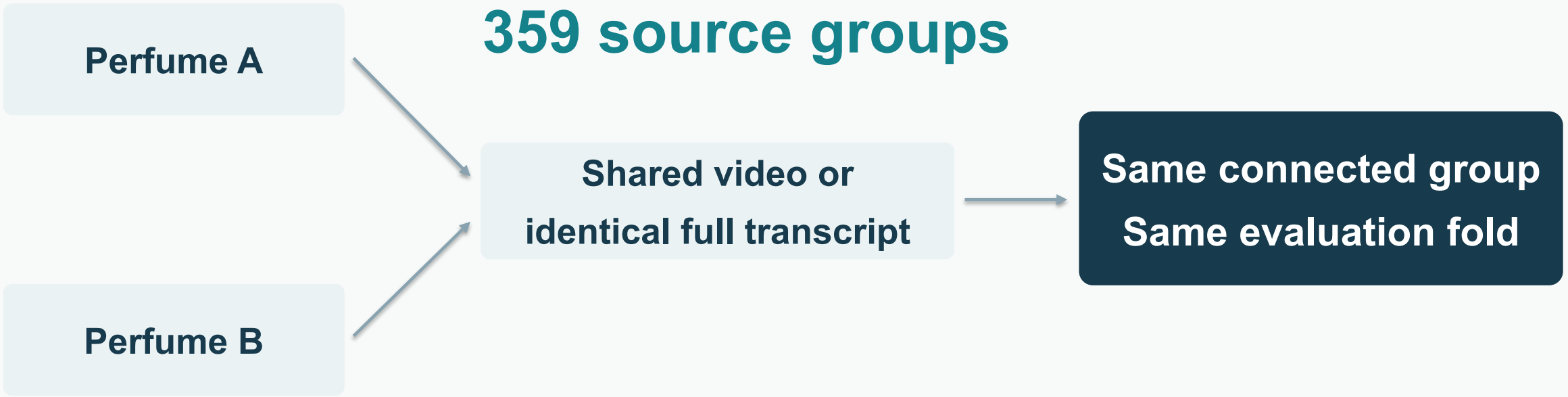
Performance is measured on held-out perfumes



Five outer folds share the same splits. Each perfume is held out for its prediction.

Test data do not guide training. Emotion labels use a fixed 0.30 probability cutoff.

Shared sources stay together



Shared videos and identical transcripts cannot cross training and test data.

Groups differ in size; supplementary analyses give each source group equal weight.

Thesis §3.7; Table 8.

Different scores answer different questions

Metric	Question
Macro Average Precision (AP)	Are perfumes with the reference emotion ranked near the top?
Macro F1	How well do final labels balance missed emotions and false alarms?
Macro ROC-AUC	How often do reference positives rank above reference negatives?

AP evaluates ranking; F1 evaluates labels at the fixed cutoff. ROC-AUC is supplementary.

“Macro” gives each emotion equal weight. These scores are not percentages correct.

Thesis §4.1.3; Saito & Rehmsmeier (2015).

A real example connects the steps

Vetiver Nocturne Absolu Extrait de Parfum

“slightly sweet medicinal herbs”

“Sweet” contributes to dictionary emotion counts.

10 transcripts → **58 features** → **Logistic Regression**

Emotion	Score	At 0.30
Joy	0.674	Present
Trust	0.604	Present
Anticipation	0.603	Present

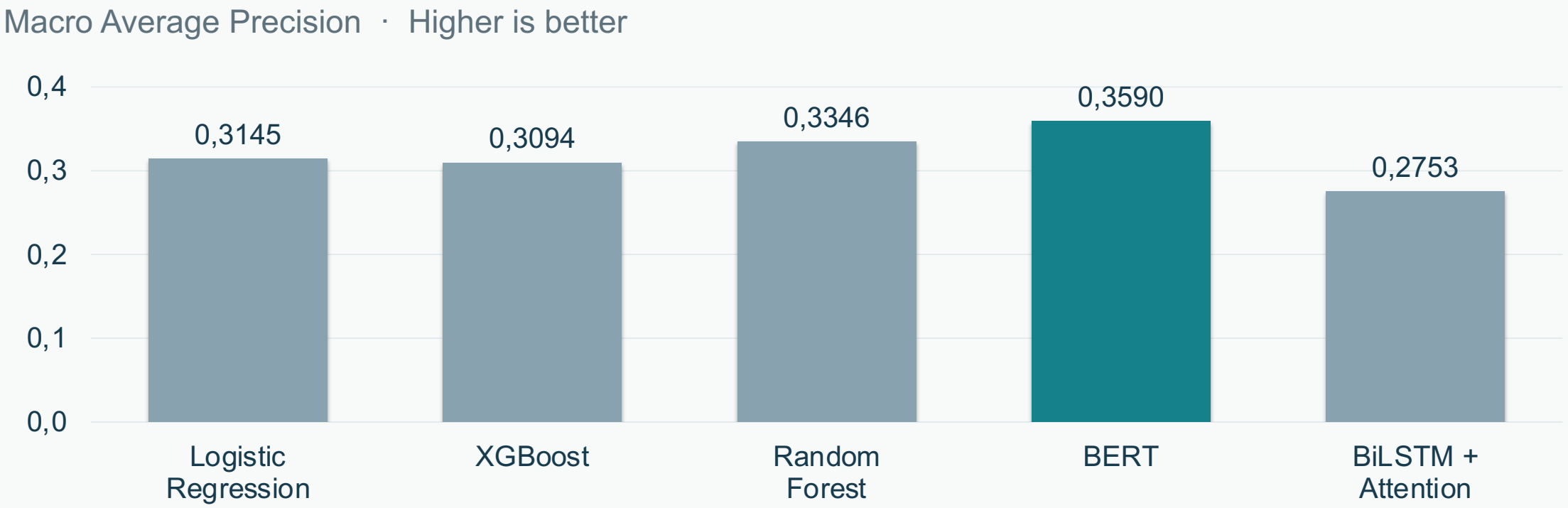
Other four scores are below the cutoff.

The result matches the reference labels.

The excerpt illustrates one feature step; prediction uses the full perfume-level representation.

Thesis §3.9; Table 7.

BERT has the highest observed Macro AP



BERT: 0.3590 Macro AP. All five models use the same 784 held-out perfumes.

YouTube-only training; pooled out-of-fold predictions. Thesis Tables 9 and 10.

The leading model depends on the metric

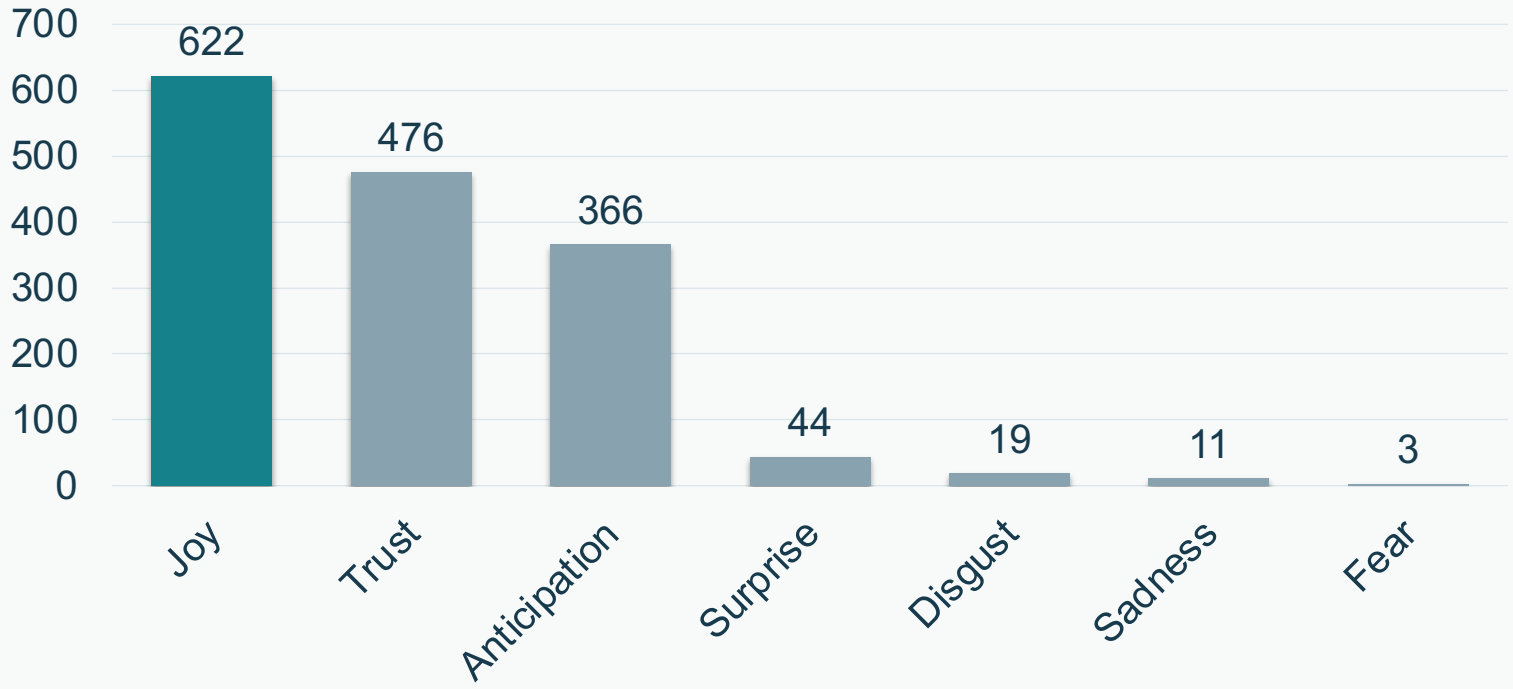
Metric	Highest pooled result	Score
Macro F1	BERT	0.4029
Macro AP	BERT	0.3590
Macro ROC-AUC	XGBoost	0.5683

With equal source-group weighting, paired F1 comparisons do not clearly separate the models.

The evidence does not establish one overall best model.

Thesis Tables 9 and 10; §4.4.2.

Emotion frequencies shape the evidence



3

positive cases for Fear

BERT's F1 of 0.5000 reflects one correct detection.

Joy, Trust and Anticipation dominate; rare-emotion scores depend on very few cases.

Thesis Table 14; §4.6.1. Counts are perfume-level reference positives.

Prediction behaviour helps explain F1

Emotion	Reference positives	Predicted positives	F1
Joy	622	784	0.8848
Trust	476	784	0.7556
Anticipation	366	784	0.6365

BERT predicts all three emotions for every perfume at the official cutoff.

Interpret F1 alongside label frequency and prediction behaviour.

Changing the cutoff changes F1; AP and ROC-AUC remain unchanged for the same probabilities.

Thesis Tables 13–15. BERT, YouTube-only training, cutoff 0.30.

Feature combinations affect models differently

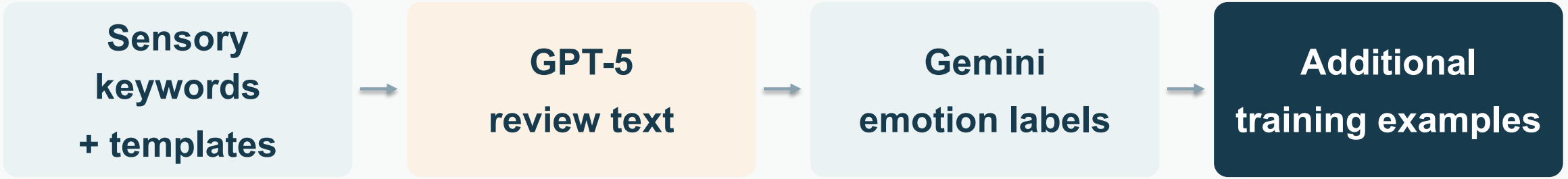
Seven feature combinations were tested across the three feature-based classifiers.

Classifier	Observed pattern
Logistic Regression	F1 changes little across feature combinations
XGBoost	Word counts + affect dimensions: F1 0.3797 Full 58-feature representation: F1 0.3260
Random Forest	The full representation has the highest observed pooled scores

The benefit of a feature family depends on the classifier.

Thesis §4.5.1; Table 12. Results use the tested fixed model settings.

Generated reviews test an additional training-data strategy

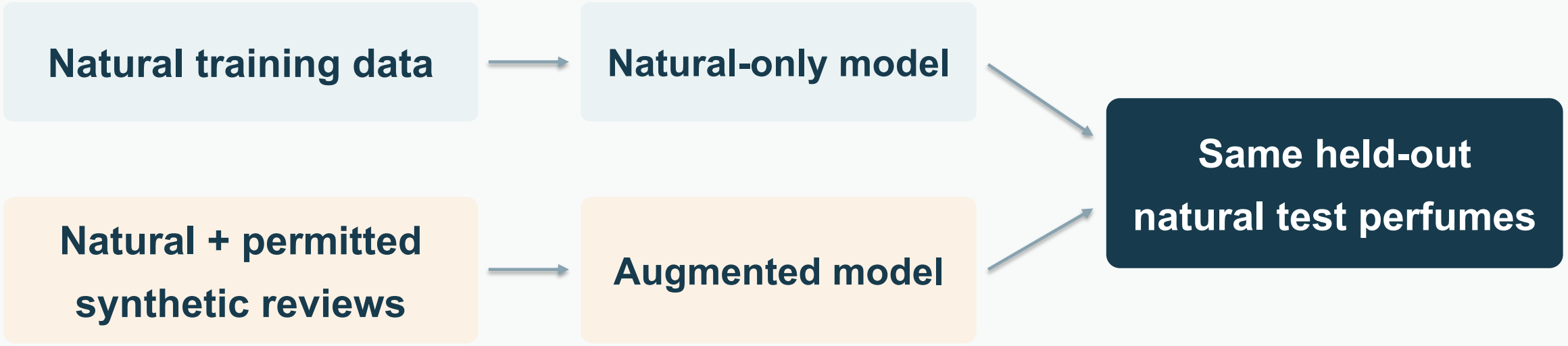


2,352 reviews in the primary corpus

Three generated reviews per perfume

Existing target emotion labels and VAD values were not supplied to GPT-5.

Both training conditions face the same natural test data



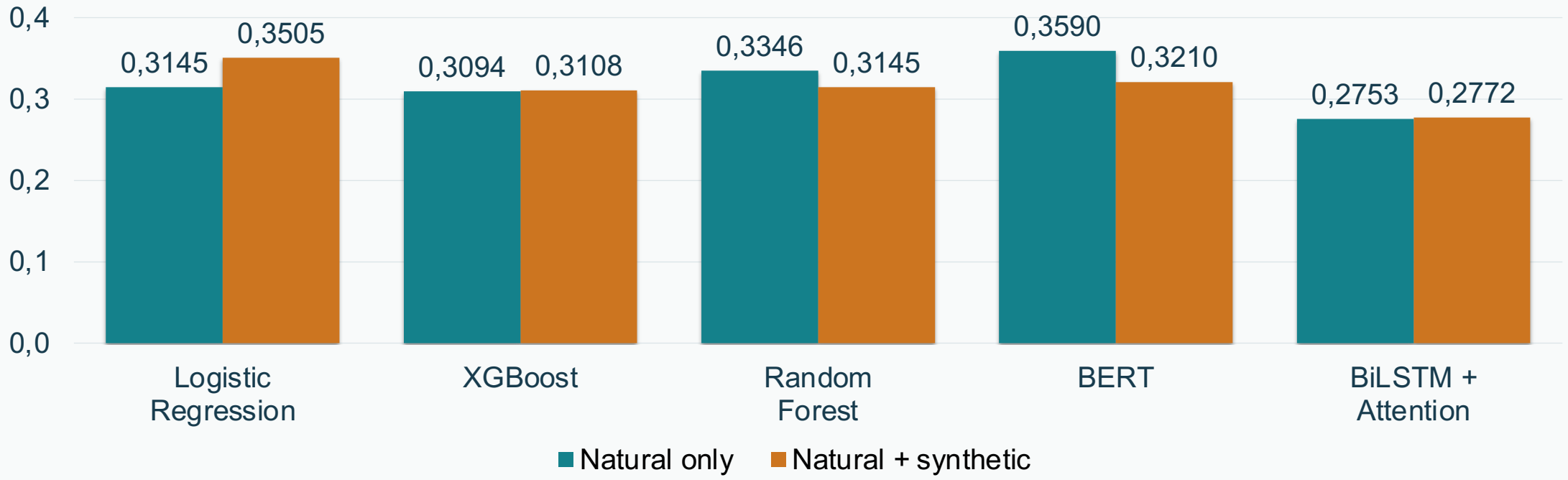
Per training perfume: natural records have total base weight 1;
the three synthetic reviews together also have total base weight 1.

Same folds, settings and cutoff. No synthetic reviews from validation or test perfumes.

Thesis §§3.8 and 4.1.1. Base weighting is before model-specific class weighting.

Augmentation produces mixed changes

Macro Average Precision · Natural-only vs. weighted augmented training



Logistic Regression improves; Random Forest and BERT decline. Other changes are small.

Thesis Tables 9 and 10. Pooled out-of-fold Macro AP on the same natural test data.

The evidence does not establish a general augmentation benefit

Changes depend on the model

The direction and size of changes differ across models and metrics.

Uncertainty remains

Under equal source-group weighting, all five 95% intervals for seven-label Macro AP change include zero.

This experiment does not isolate why the models respond differently.

Generated text needs validation on natural reviews before it can be assumed beneficial.

Thesis §4.4.3; Table 11. Paired source-group bootstrap intervals.

Answers to the research questions

RQ1

Representation

58 features combine word counts, affect dimensions and contextual scores. Their effects depend on the classifier.

RQ2

Model comparison

BERT leads pooled F1 and AP; XGBoost leads ROC-AUC. The evidence does not establish an overall best model.

RQ3

Synthetic augmentation

Weighted augmentation produces model- and metric-dependent changes, without a consistent general benefit.

The scope of the conclusions

Boundary	What it means
Model-generated reference labels	Agreement with human judgments is not established
English YouTube transcripts	Findings are specific to this review setting
Shared sources and rare emotions	Independence and estimation precision are limited

Grouping addresses shared videos and identical transcripts.

Dependence through reviewers or channels may remain.

Conclusions apply to this dataset, annotation policy and experimental configuration.

Three priorities for extending the work

1 Validate labels

Check excerpt relevance.

Collect independent human annotations.

2 Broaden the data

Add independent sources and rare-emotion cases.

Test new channels and platforms.

3 Refine augmentation

Use comparable review or perfume units.

Test generation strategies on independent natural reviews.

Three takeaways

01 Reproducible representation

Review transcripts become lexical, dimensional and contextual emotion features.

02 Comparative benchmark

Five approaches show strengths that vary across metrics and emotions.

03 Empirical augmentation finding

Generated reviews produce model-dependent changes, without consistent improvement.

A traceable dataset, a reproducible workflow and controlled evidence.

Thank you

Questions & Discussion

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784 perfumes • 5 models • 2 training conditions

Understanding emotional language in perfume reviews through reproducible modelling and evaluation.

BACKUP SLIDES →

Full results: natural training data

Model	Macro F1	Macro AP	Macro ROC-AUC
Logistic Regression	.3592	.3145	.5473
XGBoost	.3260	.3094	**.5683**
Random Forest	.3306	.3346	.5638
BERT	**.4029**	**.3590**	.5533
BiLSTM with Attention	.3486	.2753	.4507

All five models are evaluated on the same 784 held-out YouTube perfumes.

Thesis Tables 9 and 10. Pooled out-of-fold predictions; seven labels; cutoff 0.30.

Full results: augmented training data

Model	Macro F1	Macro AP	Macro ROC-AUC
Logistic Regression	**.3737**	**.3505**	.5372
XGBoost	.3359	.3108	**.5982**
Random Forest	.3247	.3145	.5731
BERT	.3316	.3210	.5267
BiLSTM with Attention	.3466	.2772	.4490

Three synthetic reviews per eligible perfume: each has base weight ⅓; natural records total 1.

Class weighting also applies. Equal base weights do not imply identical effective influence across models.

Thesis §§3.8 and 4.1.1; Tables 9 and 10. Pooled out-of-fold results; cutoff 0.30.

Source grouping and fold sizes

Outer fold	Perfumes	Source groups
1	334	1
2	113	90
3	113	90
4	112	89
5	112	89
Total	**784**	**359**

Shared videos and normalized identical full transcripts connect perfumes into groups.

Connections are transitive. Each complete group remains within one fold.

Pooled results weight perfumes equally; supplementary analyses weight source groups equally.

Thesis §3.7; Figure 5; Table 8; saved fold assignments.

Augmentation uncertainty under equal-source weighting

Model	Δ Macro AP	Unadjusted 95% interval
Logistic Regression	+.1202	[−.0366, +.1402]
XGBoost	+.0025	[−.0205, +.0381]
Random Forest	−.0244	[−.1079, +.0154]
BERT	−.1458	[−.1726, +.0722]
BiLSTM with Attention	−.0044	[−.0434, +.0274]

Seven-label Macro AP. Equal-group weighting answers a different question from pooled scores.

Across seven-label and Fear-omitted analyses, 2 of 30 intervals exclude zero.

2,000 bootstrap samples; fitted predictions held fixed; no multiple-comparison correction.

Thesis §4.4.3; Table 11. Changes are augmented minus natural-only training.

Annotation policy and validation boundaries

Fixed annotation policy

Gemini 3.1 Pro with fixed prompts.

Natural data: 1–2 excerpts, each ≤360 characters. Synthetic data: one review.

Weak or neutral evidence can produce all-zero labels.

Validation boundaries

Anger excluded: 7 natural positives and 0 synthetic positives.

174 of 784 records had no name anchor found; 20 inspected traces were correct.

Traceability does not establish topical relevance or agreement with human emotion judgments.

Thesis §3.4; Appendices 2 and 3.

Input scope and model configuration

Component	Input scope
Lexicon and affect dimensions	Full transcript
Pretrained feature models	First 512 characters
BERT	Up to 96 tokens
BiLSTM with Attention	Up to 208 tokens

BERT

GoEmotions initialization; up to 3 epochs.

BiLSTM

Trained from scratch; up to 12 epochs.

Text probabilities are averaged at perfume level. 4,879 chunks remain after removing two within-perfume duplicates.

The comparison includes differences in input scope and training configuration.

Per-emotion F1: natural training data

Emotion	Support	LR	XGBoost	RF	BERT	BiLSTM
Joy	622	.8614	.8774	.8848	.8848	.8848
Trust	476	.7376	.7179	.7556	.7556	.7556
Anticipation	366	.6380	.5831	.6294	.6365	.6365
Surprise	44	.1274	.1034	.0444	.0435	.1068
Disgust	19	.0235	.0000	.0000	.0000	.0197
Sadness	11	.0435	.0000	.0000	.0000	.0261
Fear	3	.0833	.0000	.0000	.5000	.0108

Rare-emotion scores can change substantially with one prediction.

Thesis Table 14. Perfume-level reference support; cutoff 0.30; BiLSTM uses attention.

Selected references

Nikitina & Furuoka (2023)

Words of scents: A linguistic analysis of online perfume reviews.

Nandwani & Verma (2021)

A review on sentiment analysis and emotion detection from text.

Mohammad & Turney (2013)

Crowdsourcing a Word–Emotion Association Lexicon.

Mohammad (2018)

Obtaining Reliable Human Ratings of Valence, Arousal, and Dominance for 20,000 English Words.

Demszky et al. (2020)

GoEmotions: A Dataset of Fine-Grained Emotions.

Devlin et al. (2019)

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.

Saito & Rehmsmeier (2015)

The Precision–Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets.

Niu et al. (2024)

Rethinking Emotion Annotations in the Era of Large Language Models.

All eight works are cited in the thesis. Titles are linked to the source pages.

Threshold sensitivity

Model	0.20	0.30 official	0.50	0.70
Logistic Regression	.3590	**.3592**	.2985	.1690
XGBoost	.3409	**.3260**	.2950	.2248
Random Forest	.3526	**.3306**	.2952	.1669
BERT	.3926	**.4029**	.2899	.0000
BiLSTM with Attention	.3515	**.3486**	.1943	.0000

Changing the cutoff changes the labels without retraining. AP and ROC-AUC are unchanged.

The official 0.30 cutoff was specified before this sensitivity check.

Thesis §4.5.2; Table 13. Macro F1 from the same YouTube-only out-of-fold probabilities.

Model-comparison uncertainty

Supplementary analysis: equal weight for each source group

BERT point estimate	Macro F1	Macro AP	Macro ROC-AUC
Highest observed	.4839	.4665	.5932

F1 comparisons

All 10 paired intervals include zero.

Ranking comparisons

7 of 20 intervals exclude zero, mainly involving BiLSTM.

These results do not establish one model as clearly best across all measures.

Thesis §4.4.2. Unadjusted 95% source-group bootstrap intervals; conditional on fitted models.